An Offline Intelligent Vietnamese Input Method Based on Deep Learning

Capstone Project Report 1 · AIP491

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Motivation & Problem Context

Limitations of Traditional Input Methods (IMEs)

The Input Bottleneck in Vietnamese

- Requires Latin base glyphs + **6 tonal marks** + **3 vowel modifications**.
- Rule-based systems (Telex/VNI) enforce rigid keystrokes:
  - High keystroke overhead & cognitive friction.
  - Tone corruption in English (*"case"* → *"cae"*).
  - Zero tolerance for mobile virtual keyboard typos.

Core Research Proposition

Continuous unaccented Latin typing with joint boundary segmentation and diacritic restoration.

Ergonomic Efficiency

Keystroke Savings (KS):

$$\mathrm{KS} = \left(1 - \frac{|X_{\text{continuous}}|}{|X_{\text{Telex}}|}\right) \times 100\%$$

- **20%–40% reduction** in screen taps.
- Eliminates explicit spacebar & accent keys.
- Lowers thumb fatigue & RSI risk.

Proposed Paradigm: NextKey Continuous Input

Comparison of Input Method Paradigms

Typing Paradigm	User Input Keystrokes	Restored Output	Keystrokes
Standard Telex	`t-o-o-i-space-d-d-i-space-h-o-c-j`	toi di hoc	13 taps
Standard VNI	`t-o-6-i-space-d-9-i-space-h-o-c-5`	toi di hoc	13 taps
NextKey (Proposed)	`t-o-i-d-i-h-o-c`	**toi di hoc**	**8 taps (38.5% KS)**
NextKey (With Typos)	`t-o-i-d-i-h-o-c-b-a-k-k`	**toi di hoc bai**	**12 taps ($k \rightarrow i$)**

Core Advantages

- Continuous typing flow without manual space insertion.
- Seamless multilingual typing (no English tone pollution).
- Robust against adjacent-key fat-finger errors.

Deployment Constraints

- **Inference Latency:** $<$ 50 ms per event (fluid 60fps).
- **Memory Footprint:** $<$ 2 MB INT8 model size.
- **Privacy Guarantee:** 100% offline client execution.

Literature Review & Baseline Systems

Assessment of Existing Architectural Approaches

Model Family	Representative	Strengths	Critical Gaps / Limitations
Rule-Based IMEs	Unikey, EVKey	Deterministic, zero RAM, instant speed.	Rigid; zero typo tolerance; requires explicit mode toggling for English terms.
Statistical N-Grams	HMM, Kneser-Ney	Fast inference; effective for local context.	Exponential state space; fails on unsegmented continuous sequences.
Recurrent Networks	Bi-LSTM + CRF	Captures sequential character context.	Slow autoregressive decoding; historically decoupled diacritics from typos.
Pretrained LLMs	PhoBERT, ViT5, ByT5	SOTA semantic representation.	Large footprint ($>$ 100M); high latency ($>$ 300ms); infeasible for edge.

Mathematical Problem Formulation

Formal Definition of Multi-Task Sequence Transformation

Sequence Transformation Formulation

Given noisy unaccented $X = (x_1, \dots, x_T) \in \mathcal{V}_{\text{input}}^T$, find clean target $Y = (y_1, \dots, y_M) \in \mathcal{V}_{\text{output}}^M$:

$$\hat{Y} = \arg\max_Y P(Y \mid X; \theta) = \arg\max_Y \prod_{t=1}^{M} P(y_t \mid y_{<t}, X; \theta)$$

Compound Noise Taxonomy ($\mathcal{T}$)

- **Diacritic Stripping:** Removal of all 6 tones.
- **Whitespace Erasure:** Word merging.
- **QWERTY Spatial Typos:** Neighbor keys.
- **Residual IME Codes:** Incomplete Telex.

Multi-Task Objective

$$\mathcal{L}_{\text{total}} = \lambda_{\text{diac}}\mathcal{L}_{\text{diac}} + \lambda_{\text{seg}}\mathcal{L}_{\text{seg}} + \lambda_{\text{typo}}\mathcal{L}_{\text{typo}}$$

- Joint cross-entropy optimization for diacritics, boundaries, and typos.

Core Research Questions (RQ1 – RQ6)

Foundational Scientific Inquiries Driving NextKey

Modeling & Noise Inquiries

- **RQ1 (Restoration Feasibility):** Can a compact network restore text from compound noise (accents, spaces, typos)?
- **RQ2 (Architecture Pareto Trade-off):** How do non-autoregressive taggers compare against Seq2Seq decoders in speed vs. accuracy?
- **RQ3 (Noise Calibration):** Does QWERTY spatial noise modeling improve real-world out-of-domain generalization?

Edge Optimization & UX Inquiries

- **RQ4 (Quantization & Distillation):** Can KD + INT8 quantization achieve $> 95\%$ teacher parity with < 50ms latency?
- **RQ5 (Candidate Selection Utility):** Does Top-k candidate generation improve practical typing Keystroke Savings (KS)?
- **RQ6 (Privacy-Preserving Edge):** Can on-device adaptation learn user vocabulary with zero remote server logging?

Project Objectives (S.M.A.R.T.)

Target Research & Engineering Milestones

Data & Modeling Objectives

- **Corpus Curation:** Build multi-domain parallel dataset ($>$ 840K sentences, JDWR v1) across 8 distinct genres.
- **Noise Engine:** Implement 6-dimensional QWERTY spatial and colloquial noise generator.
- **Multi-Task Learning:** Train candidate models for joint diacritic, spacing, and typo restoration.

Edge Optimization & User Targets

- **Model Compression:** Distill and quantize model to INT8 ($<$ 2 MB, $<$ 50 ms on CPU).
- **Ergonomic Savings:** Validate 20%–40% Keystroke Savings (KS) over standard Telex.
- **Zero Leakage:** Guarantee 100% offline client execution with zero network telemetry.

Project Scope & Deliverables

Clear Operational Boundaries

In-Scope Deliverables

- End-to-end deep neural sequence restoration models.
- Keyboard-aware synthetic noise generator pipeline.
- Quantized ONNX/C++ edge inference API.
- Interactive proof-of-concept web application.

Out-of-Scope Boundaries

- Low-level OS kernel keyboard drivers (iOS/Android virtual keyboard system integration).
- Voice recognition & multimodal speech input.
- Historical scripts (Hán-Nôm) or minority dialects.

Team Structure & Research Tracks

Clearly Partitioned Roles and Responsibilities

Role & Track	Assignee	Core Research & Engineering Responsibilities
Project Lead & AI Researcher	**Tran Dinh Gia** (SE181550)	• Project coordination, sprint reviews, milestone tracking. • Sequence modeling mathematical formulation & loss design. • Architecture exploration and training supervision.
Data Engineering & Preprocessing	**Truong Nhat Quang** (DE180544)	• Multi-domain corpus collection ($>$ 840K) and cleaning. • Synthetic QWERTY spatial noise & abbreviation pipeline. • In-domain and Out-of-Domain evaluation dataset curation.
Edge Systems & Deployment Lead	**Nguyen Xuan Cuong** (DE180528)	• Knowledge Distillation (KD) & INT8 quantization engines. • Edge hardware profiling and latency benchmarking ($<$ 50ms). • Offline inference backend API and interactive web interface.

Work Breakdown Structure (WBS) & Timeline

20-Week Five-Phase Execution Plan

Phase	Phase Name	Key Work Packages & Deliverables	Duration	Lead
Phase 1	**Data & Literature Foundation**	Parallel corpus collection ($>$ 840K sentences), synthetic typo & spacing noise engine, tokenization vocabulary schema.	Weeks 1–4	Quang
Phase 2	**Model Development & Baselines**	Candidate architecture training (BiGRU, T5, Transformer taggers), multi-task loss tuning, validation error analysis.	Weeks 5–9	Gia
Phase 3	**Optimization & Quantization**	Knowledge Distillation (KD) to compact Student, INT8 Post-Training Quantization (PTQ), WER/BLEU & latency benchmarking.	Weeks 10–14	Cuong
Phase 4	**Backend & Integration**	Offline C++/ONNX inference engine, REST/WebSocket keyboard API, stress testing, interactive demo prototype.	Weeks 15–17	Cuong
Phase 5	**Validation, Thesis & Defense**	Human user testing, Keystroke Savings (KS) validation, comprehensive capstone report, and defense presentation.	Weeks 18–20	All

Risk Management Framework

Risk Register and Proactive Mitigation Matrix

Identified Risk	Severity	Prob.	Proactive Mitigation Strategy
Domain Generalization Gap	High	Med	Incorporate realistic synthetic noise (QWERTY typos, chat slang, abbreviations); benchmark against isolated OOD domain (*the_thao*).
High Inference Latency	High	Med	Apply Knowledge Distillation and INT8 quantization; prioritize non-autoregressive sequence taggers over heavy autoregressive decoders.
Compute Resource Limits	Med	High	Leverage distributed dual GPU cloud allocations (Kaggle Dual T4, Google Colab); utilize mixed-precision training (FP16).
Keystroke Privacy Concerns	High	Low	Enforce 100% client-side offline execution; strictly zero remote data transmission or telemetry logging.

Quality Assurance, Evaluation & Decision Gates

Rigorous Multi-Tier Validation Methodology

Hierarchical 3-Tier Metric Harness

- **Character-Level:** CER, Typo Recovery, Boundary F1.
- **Word-Level:** WER, Word Accuracy, Token F1.
- **Sentence-Level:** Exact Match (EM), Near-Perfect ($\leq 5\%$), BLEU-4, ROUGE-L.
- **Ergonomic Metric:** Keystroke Savings (KS).

Scientific Decision Gates

- **Seed Protocol:** Fixed global seed 42 across all splits.
- **Gate 1 (Smoke Run Gate):** Reject architectures failing loss convergence within 1,000 steps.
- **Gate 2 (Objective Benchmark Gate):** Decisions strictly justified by quantitative metric gains on validation sets.
- **Budget Constraint:** $0.00 capital expenditure.

Summary of Expected Deliverables & Key Indicators

Key Performance Targets

- **Inference Latency:** $<$ 50 ms per sentence on CPU.
- **Keystroke Savings:** 20%–40% reduction vs. Telex.
- **Model Size:** $<$ 2 MB footprint via INT8 quantization.
- **Restoration Fidelity:** SOTA BLEU-4 & Word Accuracy.
- **Security:** 100% offline client-side edge inference.

Tangible Capstone Deliverables

- Open-source reproducible codebase & dataset pipeline.
- Quantized model weights & ONNX/C++ runtime.
- Interactive web demo & comprehensive thesis report.

Thank You for Your Attention!

Questions & Discussion Welcome